PyXtabond2 Methodological Note

GMM Theory, the IFE-GMM Extension, and Validation Against Stata's `xtabond2`

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Abstract

This note provides the full econometric derivation underlying `pyxtabond2`, together with a complete, primary-source-grounded account of its validation. Section 2 derives the standard dynamic panel GMM estimator (Difference and System GMM), its one- and two-step weighting matrices, the Windmeijer [2005] finite-sample variance correction, and the small-sample degrees-of-freedom corrections, each tied explicitly to the corresponding construction in Stata's `xtabond2.mata` [Roodman, 2009a]. Section 3 derives the Sargan/Hansen, Difference-in-Sargan/Hansen, and Arellano-Bond AR(ℓ) test statistics. Section 4 derives *IFE-GMM*, an original extension not present in `xtabond2`: an iterative, PCA-based estimator for dynamic panels contaminated by interactive fixed effects [Bai, 2009], combining automatic factor-count selection [Bai and Ng, 2002, Ahn and Horenstein, 2013], a profile-GMM defactoring loop [Hong et al., 2022], and a split-panel jackknife bias correction [Dhaene and Jochmans, 2015]. Section 5 reports, in full, the findings of two validation passes conducted against `xtabond2.mata` and a real Stata run, alongside every confirmed-correct result and every documented, unresolved limitation.

Contents

1 Introduction

1.1 Scope

The companion *User Manual* documents every `pyxtabond2` option practically: what it does, its Stata equivalent, and how to use it. This note instead derives *why* each computation is what it is, at the level of detail needed to audit the implementation against its econometric sources. It is organized in three parts: (i) the standard dynamic panel GMM machinery (Sections 2–3), present in Stata's `xtabond2` and reproduced by `pyxtabond2`; (ii) IFE-GMM (Section 4), the package's original extension, absent from `xtabond2`; and (iii) a full validation account (Sections 5–6) – confirmed-correct results and unresolved limitations documented rather than silently assumed away.

1.2 Notation

$i = 1, \dots, N$ indexes cross-sectional units, $t = 1, \dots, T$ time periods. y_{it} is the dependent variable, x_{it} a $p \times 1$ vector of regressors (possibly including precomputed lags of y), β the $p \times 1$ coefficient vector, η_i an unobserved time-invariant effect, ε_{it} an idiosyncratic shock. Δ denotes the first-difference operator, Z_i the instrument matrix for unit i (stacked across units as Z), W a GMM weighting matrix. H denotes a GLS-type covariance-structure matrix (Stata's `h()`). For IFE-GMM, λ_i $(r \times 1)$ is a factor-loading vector, F_t $(r \times 1)$ a common-factor vector, with $\Lambda = (\lambda_1, \dots, \lambda_N)'$ $(N \times r)$ and $F = (F_1, \dots, F_T)'$ $(T \times r)$.

2 The Standard Dynamic Panel GMM Estimator

2.1 The problem: fixed effects in a dynamic panel

Consider the workhorse dynamic panel model

$$y_{it} = \rho\, y_{i,t-1} + x_{it}'\beta + \eta_i + \varepsilon_{it}. \tag{1}$$

Because $y_{i,t-1}$ is a function of η_i, pooled OLS on (1) is inconsistent, and the within/LSDV (fixed-effects) estimator that removes η_i by demeaning is itself inconsistent for fixed T as $N \to \infty$: demeaning induces a correlation of order $O(1/T)$ between the demeaned lag and the demeaned error that does not vanish as N grows [Nickell, 1981]. Two transformations remove η_i *without* this problem, provided instruments are used for the resulting correlation between the transformed regressor and the transformed error:

First differences $\Delta y_{it} = \rho\, \Delta y_{i,t-1} + \Delta x_{it}'\beta + \Delta\varepsilon_{it}$, $t = 2, \dots, T$.

Forward orthogonal deviations (FOD) [Arellano and Bover, 1995]

$$y_{it}^{\text{FOD}} = c_{it}\left(y_{it} - \frac{1}{T_{it}^{+}}\sum_{s>t} y_{is}\right), \qquad c_{it} = \sqrt{\frac{T_{it}^{+}}{T_{it}^{+}+1}},$$

where T_{it}^{+} is the number of available future observations for unit i at t. Unlike differencing, FOD only uses *future* observations, so a single missing period does not additionally destroy the transformed values of neighbouring periods – the reason it is preferred for unbalanced panels with gaps. This is exactly `PanelData.get_fod`'s construction (`data_utils.py::_fod_group_vectorized`).

Both transformations still leave $\Delta y_{i,t-1}$ (or its FOD analogue) correlated with the transformed error, since $y_{i,t-1}$ is correlated with $\varepsilon_{i,t-1}$, which enters $\Delta\varepsilon_{it}$. This motivates instrumenting rather than direct estimation.

2.2 Moment conditions and the instrument matrix

Under sequential exogeneity/predeterminedness, $E[y_{is}\varepsilon_{it}] = 0$ for $s < t$ (more generally, for any GMM-style variable in `gmm_vars`), so that

$$E[y_{i,t-2}\,\Delta\varepsilon_{it}] = 0, \quad E[y_{i,t-3}\,\Delta\varepsilon_{it}] = 0, \ \ldots$$

for $t = 2, \ldots, T$: each time period's differenced error is orthogonal to *every* level of the instrumenting variable lagged two or more periods. This yields the classic Arellano-Bond "staircase" instrument matrix, one row block per unit [matching Windmeijer, 2005, their notation, adapted]:

$$Z_i = \begin{bmatrix} y_{i1} & 0 & 0 & 0 & 0 & 0 \\ 0 & y_{i1} & y_{i2} & 0 & 0 & 0 \\ & & \ddots & & & \\ 0 & 0 & 0 & y_{i1} & \ldots & y_{i,T-1} \end{bmatrix},$$

with $T(T-1)/2$ instrument columns for a single GMM-style variable when every lag from 1 is admissible – the maximal case, appropriate for a *predetermined* variable (uncorrelated with future, but not necessarily contemporaneous, errors). `pyxtabond2`'s own default (`lag=(2, None)`) instead starts at lag 2, the safer assumption for a genuinely *endogenous* variable such as the lagged dependent variable itself, and yields $(T-2)(T-1)/2$ columns; the construction below is identical either way, only the lag window differs. This is exactly what `SystemGMMBuilder._fill_block_a_standard` (`gmm_builder.py`) constructs: one column per (time period, lag) pair, subject to the chosen lag window. Two generalizations `pyxtabond2` implements on top of this baseline (both mirroring `xtabond2`):

- **Lag windows** (`lag_limits_diff`/`GMMStyle.lag`): restricting to lags $\ell \in [l_{\min}, l_{\max}]$ instead of "2 to maximum", reducing instrument count and often improving the Sargan/Hansen test's power [Roodman, 2009b].
- **Collapsing** (`collapse`/`GMMStyle.collapse`): summing across time periods within each lag distance, so a variable contributes $l_{\max} - l_{\min} + 1$ columns instead of $O(T^2)$. This is the single most effective lever against *instrument proliferation*: with too many instruments relative to N, GMM can overfit the endogenous variable in finite samples and the Sargan/Hansen overidentification test loses power to detect genuine misspecification [Roodman, 2009b] – a suspiciously perfect Hansen p-value is a symptom of this, not reassurance.

2.3 System GMM: adding the levels equation

Difference GMM performs poorly when y_{it} is highly persistent: lagged levels are then weak instruments for the differenced equation [Blundell and Bond, 1998]. Blundell-Bond/Arellano-Bover System GMM adds the *levels* equation

$$y_{it} = \rho\, y_{i,t-1} + x'_{it}\beta + \eta_i + \varepsilon_{it}$$

instrumented by *lagged differences* of the GMM-style variable, valid under the additional "mean stationarity" assumption $E[\Delta y_{i2}\,\eta_i] = 0$ (initial deviations from the long-run mean are uncorrelated with the fixed effect). `SystemGMMBuilder.build_system_instruments` builds, per GMM-style variable and per unit, a $2T \times n_{\text{cols}}$ block:

- Rows $1, \ldots, T$ (*differenced*-equation instruments): entry $(t, \cdot) = y_{i,t-\ell}$ for each admissible lag ℓ – Section 2.2's block.
- Rows $T+1, \ldots, 2T$ (*levels*-equation instruments): entry $(T+t, \cdot) = \Delta y_{i,t-d}$, $d = l_{\min} - 1$. In the standard (non-collapsed) style this is $T - 1$ *time-specific* columns, each isolating $\Delta y_{i,t-d}$ at exactly period t and zero elsewhere (`_fill_block_b_standard`); collapsing sums these into a single shared column (`_fill_block_b_collapse`), the levels-equation analogue of collapsing the differenced block.

The two blocks are stacked, giving Z_i a genuine $2T \times n_{\text{cols}}$ shape even though a single unit contributes at most $2T-2$ real (non-missing) rows – unused rows/instruments are naturally zeroed and, for instrument columns that are identically zero across the whole sample, dropped (`GMMEngine.__init__`'s `non_zero_cols` filter). Standard IV instruments (`iv_vars/IVStyle`, for strictly exogenous variables) are appended analogously via `build_iv_instruments`: a variable is used as its own instrument in the differenced and/or levels equation, and – following `xtabond2`'s convention exactly – appears as a *single* instrument column combining both equations whenever it instruments both, rather than as two separate columns.

2.4 One-step and two-step GMM

Stacking Y, X, Z across all units and (for System GMM) both equations, the GMM estimator for a weighting matrix W is

$$\hat{\beta}(W) = \left(X'ZWZ'X\right)^{-1} X'ZWZ'Y, \tag{2}$$

the sample analogue of minimizing the quadratic form $\left(\frac{1}{N}\sum_i Z_i'\hat{\varepsilon}_i\right)' W \left(\frac{1}{N}\sum_i Z_i'\hat{\varepsilon}_i\right)$ over β [Hansen, 1982]. **One-step** GMM ($\hat{\beta}_1$) uses a weight matrix W_1 that does *not* depend on estimated parameters – $W_1 = \left(\sum_g Z_g' H Z_g\right)^{-1}$ for a fixed H (Section 2.5); `GMMEngine._compute_W1` builds exactly this sum, group by group. **Two-step** GMM ($\hat{\beta}_2$) uses the efficient weight matrix built from the one-step residuals,

$$W_2 = \left(\sum_g Z_g'\hat{\varepsilon}_{1,g}\hat{\varepsilon}_{1,g}' Z_g\right)^{-1} = S_2^{-1}, \tag{3}$$

computed via the identity $Z_g'ee'Z_g = (Z_g'e)(Z_g'e)'$ for efficiency (`_accumulate_ZeZe`). W_2 is consistent for the inverse of the true moment covariance under heteroskedasticity/serial correlation of unrestricted form – this is what makes two-step GMM asymptotically efficient, and also what makes its *naive* standard errors unreliable in finite samples (Section 2.6).

2.5 The `h()` weighting-matrix family

W_1's H matrix models $\text{Cov}(\Delta\varepsilon_{it}, \Delta\varepsilon_{is})$ under the auxiliary assumption that ε_{it} is i.i.d. (a reasonable working assumption for constructing an efficient *one-step* weight, later made robust to violations at the two-step/robust stage). Since $\Delta\varepsilon_{it} = \varepsilon_{it} - \varepsilon_{i,t-1}$,

$$\text{Cov}(\Delta\varepsilon_{it}, \Delta\varepsilon_{is}) = \begin{cases} 2\sigma^2 & s = t \\ -\sigma^2 & |s-t| = 1 \\ 0 & |s-t| \geq 2, \end{cases}$$

giving a tridiagonal H with 2 on the diagonal and -1 on the first off-diagonals. This is exactly `_build_H_diff_block`'s construction – matching the matrix Windmeijer's own paper describes for the differenced-equation weight ("2's on the main diagonal, -1's on the first off-diagonals"). Stata's `h()` option (`h`, default 3) selects among three variants, generalized in `pyxtabond2` by `_build_H(h, T_span, orthogonal)`:

`h=1` $H = I$: assumes no serial-correlation structure at all in the transformed error, i.e. does not model the covariance induced by differencing/FOD itself. Rarely appropriate, but available for comparability with software that defaults to it.

`h=2` The tridiagonal H above for the differenced block; for System GMM, the cross-covariance between the differenced and levels blocks is assumed zero (block-diagonal H).

h=3 (Stata's default) As h=2, but additionally models the *true* cross-covariance between the differenced and levels equations. Since $\mathrm{Cov}(\Delta\varepsilon_{it}, \varepsilon_{is}) = \sigma^2$ if $s = t$ and $-\sigma^2$ if $s = t-1$ (zero otherwise), the cross-block has $+1$ on its main diagonal and -1 on one off-diagonal – exactly `_build_H_full_system`'s `H_full[i,T_span+i]=1.0` / `H_full[i2+1,T_span+i2]=-1.0` assignments.

`h=2` and `h=3` coincide for pure Difference GMM (no levels block exists to model a cross-covariance with). The FOD case uses an analogous but distinct construction (`_xform_matrix`), reflecting FOD's own weights rather than the simple differencing weights, and not reproduced in full here.

2.6 Standard errors and the Windmeijer (2005) correction

2.6.1 One-step

Non-robust one-step variance is $V_1 = (X'ZW_1Z'X)^{-1}$, scaled by an estimated error variance $\hat{\sigma}^2$ computed from the *same* equation H models (Section 3.1.1 details exactly which residuals enter $\hat{\sigma}^2$ and why this depends on `h`). Robust one-step variance uses the sandwich

$$V_{1,\text{rob}} = V_1 \left(X'ZW_1\right) S_2 \left(W_1Z'X\right) V_1,$$

with S_2 as in (3) – `estimator.py`'s one-step-robust branch. Additionally, when `robust=True` with one-step estimation, `pyxtabond2` replicates `xtabond2`'s "Roodman's trick": a silent two-step refit computed *only* to obtain a valid Hansen statistic (the non-robust Sargan statistic is otherwise the only overidentification test available at one-step), without altering the reported one-step $\hat{\beta}$/se.

2.6.2 Two-step: why naive standard errors are wrong

W_2 in (3) is itself a function of $\hat{\beta}_1$ (through $\hat{\varepsilon}_1$). The naive two-step variance $V_2 = (X'ZW_2Z'X)^{-1}$ treats W_2 as if it were the *true, non-stochastic* W, ignoring the extra sampling variability it inherits from $\hat{\beta}_1$. Windmeijer [2005] shows this omission is the dominant source of the well-documented downward bias of two-step GMM standard errors in finite samples. Writing the moment vector as $\bar{g}(\beta) = \frac{1}{N}\sum_i g_i(\beta)$ and the efficient weight as $W_N(\hat{\beta}_1) = \frac{1}{N}\sum_i g_i(\hat{\beta}_1)g_i(\hat{\beta}_1)'$, a first-order Taylor expansion of $\hat{\beta}_2$ around the truth β_0, *additionally* expanding $\hat{\beta}_1$ around β_0, gives [Windmeijer, 2005, eq. 2.3–2.4]

$$\hat{\beta}_2 - \beta_0 = -A^{-1}_{\beta_0,W_N(\beta_0)} b_{\beta_0,W_N(\beta_0)} + D_{\beta_0,W_N(\beta_0)} \left(\hat{\beta}_1 - \beta_0\right) + O_p(N^{-3/2}),$$

In general D's j-th column has three terms [Windmeijer, 2005, eq. 2.3], one of which involves $G(\beta_0) = \partial^2\bar{g}(\beta)/\partial\beta\partial\beta'|_{\beta_0}$ and vanishes whenever $C(\beta) = \partial\bar{g}(\beta)/\partial\beta'$ is constant in β – exactly `pyxtabond2`'s case, since its moment conditions are *linear* ($g_i(\beta) = Z_i'(Y_i - X_i\beta)$, so $C(\beta) = -\frac{1}{N}\sum_i Z_i'X_i$ regardless of β). Writing, for brevity,

$$K_j(\beta_0) \equiv A^{-1}C(\beta_0)'W_N^{-1}(\beta_0)\, \frac{\partial W_N(\beta)}{\partial \beta_j}\bigg|_{\beta_0},$$

the two remaining terms give

$$D_{\cdot j} = -K_j(\beta_0)\, W_N^{-1}(\beta_0)\, C(\beta_0)A^{-1}b_{\beta_0,W_N(\beta_0)} \;+\; K_j(\beta_0)\, W_N^{-1}(\beta_0)\, \bar{g}(\beta_0).$$

This linear-moment-conditions case is precisely the one Windmeijer [2005] shows the correction *unambiguously* improves finite-sample accuracy (his eq. 3.1–3.2, specialized to a panel AR(1)-type model). The corrected variance is

$$\widehat{\text{var}}_c(\hat{\beta}_2) = V_2 + D\,\widehat{\text{var}}(\hat{\beta}_1)\,D' + D\,V_2 + V_2\,D'. \tag{4}$$

`GMMEngine.estimate_two_step_robust` implements exactly (4): `V1_robust_windmeijer` plays the role of $\widehat{\mathrm{var}}(\hat{\beta}_1)$ (the first-step's own sandwich variance, accumulated per cluster consistently with clustering elsewhere in the engine); `D_sum` accumulates $\partial W_N(\beta)/\partial\beta_j|_{\hat{\beta}_1}$'s two-cluster-sum components $(-Z_i' X_{i,j} \cdot (Z_i'\hat{\varepsilon}_1)'$ and its transpose, per the linear-moment-condition simplification above); and `V2_robust` assembles (4) directly. This is the variance actually reported whenever `twostep=True, robust=True` – the combination most commonly used in applied dynamic-panel work, and the only one for which naive two-step inference is known to be seriously unreliable.

2.7 Small-sample degrees-of-freedom corrections

When `small=True`, `pyxtabond2` rescales variances by finite-sample "qc" factors, matching `xtabond2.mata` lines 562–566 exactly:

$$V \leftarrow V \times \begin{cases} \dfrac{N_{\text{obs}}}{N_{\text{obs}} - k} & \text{one-step, non-robust} \\ \dfrac{N_{\text{obs}} - 1}{N_{\text{obs}} - k} \times q_{\text{cl}} & \text{one-step-robust, two-step (robust or not)} \end{cases}$$

where k is the regressor count (net of the IFE-GMM incidental-parameter penalty, Section 4.10), and

$$q_{\text{cl}} = \begin{cases} 1 & \texttt{cluster()} \text{ induces a partition different from the panel id} \\ \dfrac{N_{\text{clusters}}}{N_{\text{clusters}} - 1} & \text{clustering induces the same partition as the panel id} \end{cases}$$

– `GMMEngine._cluster_qc_factor`. The second case covers both the default (no `cluster()` given, so clustering falls back to the panel id) and an explicit `cluster()` that happens to induce the same partition as the panel id (e.g. `cluster(country_id)` on a panel already indexed by `country_id`): both leave the moment structure identical to the no-`cluster()` case, so both receive the same correction factor. Only a `cluster()` that genuinely induces a different grouping than the panel id drops the factor. This asymmetry (Mata's `rows(clusts)` conditional) is confirmed directly from `xtabond2.mata` line 564. N_{obs} itself is the levels-equation observation count for System GMM (matching Stata's own `NObs`, computed from the levels rows) and the differenced-equation count for Difference GMM.

3 Diagnostic Tests

3.1 Sargan and Hansen overidentification tests

The (non-robust) Sargan statistic tests H_0: all instruments are valid,

$$\text{Sargan} = \frac{\hat{\varepsilon}_1' Z\, W_1\, Z' \hat{\varepsilon}_1}{\hat{\sigma}^2} \;\sim\; \chi^2_{n_{\text{instr}} - k},$$

valid only under the same homoskedasticity/no-serial-correlation auxiliary assumption used to build W_1. The Hansen statistic, $\hat{\varepsilon}_2' Z\, W_2\, Z' \hat{\varepsilon}_2 \sim \chi^2_{n_{\text{instr}} - k}$, is consistent under the actual (possibly heteroskedastic, clustered, serially correlated) moment covariance and should be preferred whenever available. Both lose power as instruments proliferate [Roodman, 2009b]: with many instruments, both statistics can fail to reject even a misspecified model, so a very high p-value (e.g. > 0.99) signals over-instrumentation rather than validity.

3.1.1 The error-variance scale $\hat{\sigma}^2$

`xtabond2.mata:201` defines `ErrorEq`, "a dummy for [the] equation whose errors to use for sig2 ... transformed eq unless $h = 1$" – i.e. $\hat{\sigma}^2$ is computed from the transformed (differenced/FOD) equation's residuals when $h > 1$, but from the *levels*-equation residuals when $h = 1$ (consistent with `h=1`'s own assumption that the transformed error itself has no differencing-induced structure):

$$\hat{\sigma}^2 = \frac{1}{d} \cdot \frac{1}{N_{\text{eq}}} \sum \hat{\varepsilon}^2_{1,\text{eq}}, \qquad (\text{eq}, d) = \begin{cases} (\text{levels},\ 1) & h = 1 \\ (\text{diff/FOD},\ 1) & \text{FOD} \\ (\text{diff},\ 2) & \text{otherwise.} \end{cases}$$

This is exactly `get_diagnostics`'s branch on `self.h`.

3.2 Difference-in-Sargan/Hansen tests

Instrument *subset* exogeneity is tested by refitting with that subset excluded and comparing overidentification statistics:

$$\text{Diff} = \text{Stat}_{\text{full}} - \text{Stat}_{\text{restricted}} \ \sim\ \chi^2_{n_{\text{instr,full}} - n_{\text{instr,restricted}}}$$

under H_0: the excluded subset is valid. `pyxtabond2` tests two subsets automatically, mirroring `xtabond2`: the System GMM "GMM instruments for levels" block (identified as the columns that are zero throughout the differenced rows and non-constant on the levels rows – excluding the automatically-added constant instrument), and a second block covering all `iv()`/`iv_vars` instruments jointly – every standard-instrument variable, across every `IVStyle` group, tested together as a single combined subset rather than group by group. `diagnostics.py`'s `compute_diff_sargan_tests` refits a fresh `GMMEngine` on the restricted instrument set with `check_rank=False` (the full model was already identification-checked; a diagnostic refit on a column *subset* of an already-validated Z does not need a second full-Z rank SVD).

3.3 Arellano-Bond AR(ℓ) tests

The AR(ℓ) test targets H_0: no ℓ-th order serial correlation in the *original* idiosyncratic error ε (not the transformed one), using $w_{it} = \hat{\varepsilon}_{i,t-\ell}$ (the transformed-equation residual lagged ℓ periods) and the raw moment $m_\ell = \sum_{i,t} w_{it}\hat{\varepsilon}_{it}$. Because $\hat{\varepsilon}$ depends on the estimated $\hat{\beta}$ rather than the truth, a naive variance for m_ℓ is invalid; the correctly propagated variance (matching `xtabond2.mata::_ARTests`, implemented in `GMMEngine.get_diagnostics::compute_ar`) is

$$v_\ell = w'Hw + (X'w)' \left[-2VZXA \cdot (Z'Hw) + V_{\text{rob}} \cdot (X'w)\right], \qquad z_\ell = \frac{m_\ell}{\sqrt{v_\ell}} \overset{a}{\sim} N(0,1),$$

where $VZXA = V \cdot X'Z \cdot W$ is the GMM "bread" and V_{rob}/V are the (possibly robust) parameter covariance matrices – the extra terms propagate the estimation uncertainty in $\hat{\beta}$ into the test. By construction, $\Delta\varepsilon_{it} = \varepsilon_{it} - \varepsilon_{i,t-1}$ is first-order correlated *even under the null of no correlation in* ε ($\text{Cov}(\Delta\varepsilon_{it}, \Delta\varepsilon_{i,t-1}) = -\text{Var}(\varepsilon_{i,t-1}) \neq 0$), so **rejecting AR(1) is the expected, uninformative outcome**; failing to reject AR(2) or higher is the actual validity requirement for lag-2+ instruments (a rejected AR(2) means the original ε was itself serially correlated, invalidating those instruments). `artests` generalizes ℓ beyond Stata's hardcoded AR(1)/AR(2) to any $\ell = 1, \ldots, \ell_{\max}$ (set via `artests`). `arlevels` redirects the test to the *levels*-equation residuals (System GMM only), forcing H's $w'Hw$ term to the identity ($h = 1$-style) regardless of the model's own `h` – matching `xtabond2.mata`'s `H = _H(arlevels?1:h, ...)` – while the $Z'Hw$ cross-term is *not* forced (Mata's `psit = _H(h,...)` always uses the actual `h`).

3.4 Wald/F test

Joint significance of all regressors uses Wald $= \hat{\beta}'V^{-1}\hat{\beta}$ (constant excluded, via a pseudo-inverse guarding against near-singular V), reported as Wald$/k_{\text{indep}} \sim F_{k_{\text{indep}}, \text{df}_{\text{resid}}}$ under `small=True` or Wald $\sim \chi^2_{k_{\text{indep}}}$ otherwise.

4 IFE-GMM: An Original Extension for Interactive Fixed Effects

4.1 Motivation

Sections 2–3 assume ε_{it} is idiosyncratic conditional on η_i. If the true error instead has an *interactive fixed effects* (multifactor) structure,

$$u_{it} = \lambda_i' F_t + \varepsilon_{it}, \tag{5}$$

with λ_i a unit-specific loading on a common, possibly time-correlated shock F_t [Bai, 2009], and if F_t is correlated with the regressors (a common macro shock driving both x_{it} and y_{it}, for instance), then $\lambda_i' F_t$ acts as an omitted, regressor-correlated term that first-differencing/FOD does *not* remove (differencing eliminates the time-*invariant* η_i, not the time-*varying* F_t). Standard GMM applied to (1) in the presence of (5) is therefore inconsistent. Note that (5) *nests* the additive fixed-effects case as the special case $r = 1$, $\lambda_i \equiv 1$: it is a strict generalization, not a different model class [Bai, 2009, Section 1].

4.2 Identification: the r^2 restrictions

$\Lambda F' = \Lambda A A^{-1} F'$ for any invertible $r \times r$ A, so (Λ, F) is unidentified without normalizing restrictions. Since an invertible $r \times r$ matrix has r^2 free elements, exactly r^2 restrictions are needed [Bai, 2009, Section 3.1]. Two standard sets jointly deliver this and uniquely determine (Λ, F) given the product $\Lambda F'$:

$$F'F/T = I_r \quad (r(r{+}1)/2 \text{ restrictions, symmetry}), \qquad \Lambda'\Lambda = \text{diagonal} \quad (r(r{-}1)/2 \text{ restrictions}),$$

summing to $r(r + 1)/2 + r(r - 1)/2 = r^2$ exactly. `pyxtabond2`'s IFE-GMM loop (`ife.py`) enforces $F'F/T = I_r$ directly by construction (factors are extracted as the top-r right singular vectors of the residual matrix, rescaled by $\sqrt{T}$: `F = Vt[:r,:].T * sqrt(n_times)`), which is sufficient for its purposes – the estimator's target is β, for which only the *column space* of F (not the individual loadings) matters, so the $\Lambda'\Lambda$-diagonal restriction is not separately imposed.

4.3 Defactoring: the projection matrix M_F

Given β, the residual $W_i = Y_i - X_i\beta$ follows the pure factor model $W_i = F\lambda_i + \varepsilon_i$ [Bai, 2009, eq. 9]. Concentrating out $\Lambda = W'F/T$ from the least-squares objective yields, for a candidate F, the projection

$$M_F = I_T - F(F'F)^{-1}F' = I_T - FF'/T, \tag{6}$$

[Bai, 2009, Section 3.2] with $M_F W_i = M_F \varepsilon_i$ (the factor component is exactly annihilated). F itself is recovered as the top-r eigenvectors (scaled by $\sqrt{T}$) of $\sum_i W_i W_i' = WW'$, i.e. via SVD of the residual matrix – exactly `ife.py`'s `U, S, Vt = svd(E_mat); F = Vt[:r,:].T * sqrt(n_times)`, and (6) is exactly `M_F = np.eye(n_times) - (F @ F.T) / n_times`. Bai's own baseline iteration alternates $\hat{\beta}(F, \Lambda) = (\sum_i X_i' X_i)^{-1} \sum_i X_i'(Y_i - F\lambda_i)$ (ordinary least squares, given the factor structure) with re-extracting (F, Λ) from the implied pure-factor residuals.

4.4 From least squares to profile GMM

Bai's baseline update for β is OLS, valid when X_{it} is (conditional on the factors) exogenous. In a *dynamic* panel, X_{it} includes $y_{i,t-1}$, mechanically endogenous even after defactoring – OLS on defactored data would reproduce the Nickell-type bias (Section 2) all over again. Hong et al. [2022] generalize Bai's scheme by replacing the OLS update with a GMM update using *defactored instruments*: given F (hence M_F), project Y, X, *and* Z onto M_F's column space and re-run standard GMM (Section 2.4) on the projected data. They note (their eq. 2.8) that substituting the *estimated* loadings $\hat{\lambda}_i$ directly into a per-unit defactoring is a tempting "naive" alternative, but complicates the asymptotic analysis; their recommended construction (their eq. 2.9) is instead to defactor with the *projection* M_F applied uniformly, exactly as in (6) – precisely `ife.py`'s choice: `Y_proj = Y_mat @ M_F`, and likewise for every column of X and every GMM/IV-instrumented variable, *before* the GMM engine ever sees the data for that iteration.

4.5 The iterative algorithm

1. **Initialize.** $\hat{\beta}^{(0)} \leftarrow$ plain GMM on the raw (non-defactored) data (Section 2.4).
2. **Residuals.** $\hat{u}_{it}^{(m)} = y_{it} - x_{it}'\hat{\beta}^{(m)}$, reshaped into an $N \times T$ matrix $\hat{U}^{(m)}$.
3. **Factors.** $F^{(m)} \leftarrow \sqrt{T}\times$(top-$r$ right singular vectors of $\hat{U}^{(m)}$); $M_F^{(m)}$ via (6).
4. **Project.** $\tilde{Y}^{(m)} = Y M_F^{(m)}$, $\tilde{X}_{\cdot,j}^{(m)} = X_{\cdot,j} M_F^{(m)}$ for every regressor j, and likewise for every GMM-style/IV-style variable feeding the instrument matrix.
5. **Re-estimate.** $\hat{\beta}^{(m+1)} \leftarrow$ GMM on $(\tilde{Y}^{(m)}, \tilde{X}^{(m)}, \tilde{Z}^{(m)})$.
6. **Converge?** If $\sum_j |\hat{\beta}_j^{(m+1)} - \hat{\beta}_j^{(m)}| < \tau$ (`ife_tol`, default 10^{-5}), stop; else repeat from step 2, up to `ife_max_iter` (default 30) iterations.

Diagnostics (Section 3) are computed only once, on the final converged fit – they are unnecessary at intermediate iterations, where only $\hat{\beta}^{(m)}$ feeds the next step (an optimization exploited in the package's performance work, orthogonal to the econometrics documented here).

Missing panel cells. $\hat{U}^{(m)}$ can contain undefined entries for two distinct reasons that leave the same kind of hole: a genuinely unbalanced panel (an (i,t) observation absent from the data), or simply a regressor's lag-truncated first observation per unit (present in the data, but with an undefined residual). Rather than a separate imputation pass, each such cell is filled from the *previous* iteration's own rank-r reconstruction of $\hat{U}$ – $U_{\cdot,1:r}\,\mathrm{diag}(S_{1:r})\,V_{\cdot,1:r}'$ from that iteration's SVD, or the cross-sectional mean on the very first iteration, before any reconstruction exists – before the SVD in step 3 above. The outer loop already alternates between a factor-structure estimate and a data reconstruction the same way an EM algorithm would, so no nested imputation loop is needed. `pyxtabond2` prints a one-time, informational console notice the first time this triggers per fit; it does not alter the structure of the returned result (Section 6 discusses the caveat this raises under heavy missingness).

4.6 Convergence

Hong et al. [2022] prove a *contraction mapping* property for exactly the update in Section 4.5: iterating converges quickly to the true β *provided the starting value* $\hat{\beta}^{(0)}$ *is already consistent under interactive fixed effects.* They obtain such a starting value via a dedicated nuclear-norm-regularized (NNR) estimator. `pyxtabond2`'s loop instead starts from plain, non-defactored GMM (Section 4.5, step 1) – *not* guaranteed consistent when factors are strong, since plain GMM is exactly the estimator IFE-GMM exists to correct. Citing the iterative estimator they compare against [Jiang et al. 2021, as discussed in Hong et al., 2022]: "the asymptotic properties of ... iterative estimators depend on the initial estimator, the population structure and the iterative steps, and they may not be consistent if the initial estimators are inconsistent." `pyxtabond2` does not implement an NNR (or other factor-consistent) starting point; empirical convergence should be read as a favourable sign, not a proof of correctness, particularly when factor loadings are large relative to ε's scale.

4.7 Factor-number selection

When `r='auto'` is used, `pyxtabond2`'s factor-count routine (`numfac.estimate_num_factors`, a direct port of Stata's `xtnumfac.ado`) computes, from the plain-GMM residual matrix E ($N \times T$, or its EM-PCA imputation if unbalanced) and its singular values $S_1 \geq \cdots \geq S_{\min(N,T)}$, $\mu_j = S_j^2/(NT)$:

Bai & Ng (2002) IC_{p2}. With $V(k) = \frac{1}{NT}\sum_{j>k}\mu_j$ (residual variance after removing k factors) and $V(0) = \overline{E^2}$,

$$IC_{p2}(k) = \ln V(k) + k \cdot \frac{N+T}{NT}\ln\big(\min(N,T)\big), \qquad \hat{r}_{IC} = \arg\min_{k\geq 1} IC_{p2}(k).$$

Ahn & Horenstein (2013) eigenvalue ratio (ER).

$$ER(k) = \frac{\mu_{k-1}}{\mu_k} \quad (k \geq 1), \qquad ER(0) = \frac{V(0)/\ln(\min(N,T))}{\mu_1} \quad \text{(the "mockEV" device)},$$

$$\hat{r}_{ER} = \arg\max_{k\geq 0}\ ER(k).$$

The `mockEV` term stands in for a "zeroth eigenvalue", the specific device that lets $\hat{r}_{ER} = 0$ be a reachable outcome – concluding "no factor structure" – which IC_{p2} cannot do on its own. `pyxtabond2` reports $\hat{r}_{ER}$ as the default automatic choice (`fac_results['best_er']`), alongside IC_{p2} for comparison, matching `xtnumfac`'s own reported preference.

4.8 Split-panel jackknife bias correction

Any estimator with an $O(1/T)$ asymptotic bias term, $\hat{\beta}(T) = \beta_0 + B/T + o_p(1/T)$, can be bias-corrected by exploiting how that bias scales with T: splitting the sample into two halves of length $T/2$ gives $\hat{\beta}(T/2) = \beta_0 + 2B/T + o_p(1/T)$ for each half, so

$$2\hat{\beta}(T) - \tfrac{1}{2}\left(\hat{\beta}_{t\leq\bar{t}}(T/2) + \hat{\beta}_{t>\bar{t}}(T/2)\right) = \beta_0 + \left(2\cdot\tfrac{B}{T} - \tfrac{1}{2}\cdot 2\cdot\tfrac{2B}{T}\right) + o_p(1/T) = \beta_0 + o_p(1/T),$$

cancelling the leading bias term exactly – the classical Quenouille/Tukey jackknife idea, applied along the *time* dimension (appropriate here since the bias is specifically $O(1/T)$, an incidental-parameter-style artifact of estimating T-growing factor structure alongside β, not an $O(1/N)$ phenomenon). Dhaene and Jochmans [2015] formalize exactly this split-panel jackknife for fixed-effect models with an incidental-parameter bias, with the $g = 2$ case of their general g-way-split estimator (their eq. 2.3) being precisely

$$\hat{\beta}_{\mathrm{BC}} = 2\hat{\beta}_{\mathrm{full}} - \tfrac{1}{2}\left(\hat{\beta}_{t\leq\bar{t}} + \hat{\beta}_{t>\bar{t}}\right), \tag{7}$$

matching `ife.py`'s `bias_correction=True` computation verbatim, with $\bar{t}$ the midpoint time period, and each half-panel estimate obtained by re-running the *entire* IFE-GMM loop (Section 4.5) on that half, warm-started from the full-panel $\hat{\beta}$ (`beta_warm_start`) to avoid re-solving from a cold start on a shorter, noisier sub-panel.

4.9 Asymptotic theory: variance vs. bias

Hong et al. [2022, Theorem 3.4] characterize both the asymptotic variance and bias of exactly the iterative estimator of Section 4.5. Two results matter directly for `pyxtabond2`'s design:

1. The estimator's asymptotic *variance* coincides, to leading order, with plain GMM's variance evaluated at the (unknown) true factors – no additional term for the sampling uncertainty of the *estimated* factors is needed. This validates `pyxtabond2`'s practice of reporting the plain GMM sandwich/Windmeijer-corrected variance (Section 2.6) computed on the *converged, projected* data, with no IFE-specific variance adjustment.
2. The estimator's asymptotic *bias*, in contrast, has three non-zero terms in general (analogous to Bai's own B/C bias terms for the additive-effects incidental-parameter problem), vanishing only under special cases: strictly exogenous instruments, no cross-sectional heteroskedasticity, and no serial correlation/heteroskedasticity in ε_{it}. These do not hold in general dynamic-panel applications (the entire point of instrumenting is to accommodate endogenous/predetermined regressors).

Together, these results relocate the practical risk of using IFE-GMM without bias correction (i.e. leaving `bias_correction=True` unset): it is a risk to the *point estimate*, not (to leading order) to the reported standard error – exactly the split-panel jackknife of Section 4.8's target. `bias_correction=True` is therefore recommended whenever point estimates (not just significance tests) are of interest, at roughly double the computational cost (two additional half-panel IFE-GMM loops).

4.10 Incidental-parameter degrees-of-freedom penalty

Estimating r factors and $N + T$ loadings/factors alongside β consumes degrees of freedom beyond k (the regressor count). `pyxtabond2` applies the penalty `df_pca` $= r(N_{\text{groups}} + T_{\text{span}} - r)$ wherever Stata's own small-sample corrections subtract k (Sections 2.7, 3.3), matching the standard incidental-parameter count for (5) under the Section 4.2 restrictions: $N \times r + T \times r$ loading/factor elements, less the r^2 identifying restrictions – i.e. exactly the parameter count implied by Bai [2009, p. 1235]'s own identification argument, re-derived in Section 4.2.

5 Validation Methodology

Two independent evidence sources underlie every claim in this note:

1. **Line-by-line audit against** `xtabond2.mata` (and, for IFE-GMM's factor-selection step, `xtnumfac.ado`) – the actual Mata source of Stata's `xtabond2/xtnumfac` commands (the `.ado` files are thin dispatch wrappers; all matrix algebra lives in `.mata`), read directly rather than inferred from documentation.
2. **A real Stata run** of a consolidated do-file, with `xtabond2` actually installed, comparing its log line-by-line, coefficient-by-coefficient and test-by-test, against a frozen Python reference output (both live in `stata_validation/` in the project's development repository – not necessarily included in every source distribution).

A third, complementary safety net – `tests/test_reference_values.py`'s frozen regression fixtures (development repository) – does not independently establish Stata-fidelity (it has no external ground truth) but does guarantee that any change to the estimator does not silently alter any *other* already-validated computation; every change was followed by a full `pytest` pass before being considered complete.

5.1 Unbalanced panels: validation status

Both the standard GMM engine and IFE-GMM support panels with missing (i, t) rows – interior gaps, late-starting units, and early attrition – mirroring `xtabond2.mata`'s own dense-grid (`Fill`) handling of unbalanced data, rather than requiring the caller to pre-densify the panel. Two scenarios were cross-validated against a real Stata run, matching to 6 decimal places on every coefficient and standard error: an interior gap, and a late-starting individual. The latter was chosen specifically because it is the one scenario able to distinguish between two readings of the level equation's sample-selection rule – relative to the panel's *global* $t_{\min}$, or to each unit's *own* first observed period – that are indistinguishable on every other validated specification (every unit there starts at the panel's global $t_{\min}$). The match confirms the already-implemented "global" reading is the one Stata itself uses; no change was needed. End-of-panel attrition was validated internally (finite, well-posed inference, across all three AR-test weighting branches of Section 3.3) but has not yet been independently re-run against Stata for that specific boundary case (Section 6).

6 Known Limitations

Weights (`pweight/aweight/fweight`) Not implemented. `xtabond2.mata` shows weights entering at least four distinct, non-trivial places – the W_1 accumulation, the error-variance scale (via a `pwe = e1:/sqrt(wt)` device, not a simple pre-multiplication), the two-step "meat", and the AR tests – and only the first was verified with the same rigor as the rest of this note before the decision was made not to ship a partially-verified feature.

`cluster()`: single variable only Stata's multi-way `cluster(var1 var2)` (Cameron-Gelbach-Miller alternating-sum clustering) is out of scope; most macro panels cluster on the panel id itself in any case.

`cluster()` + `twostep/robust`, instruments > clusters S_2 (eq. 3) becomes exactly rank-deficient (rank $\leq N_{\text{clusters}}$). NumPy's Moore-Penrose pseudo-inverse and Mata's `invsym`-based generalized inverse are both *valid* generalized inverses of a singular matrix, but not numerically the *same* one – point estimates (not just SE) can diverge from Stata in this specific regime (confirmed: away from it, e.g. `cluster()` on the panel id itself, results match exactly). Mitigation: keep instrument count $\leq$ cluster count (e.g. via `collapse=True`) – the same mitigation Stata's own on-screen warning for this situation implies.

`GMMStyle/IVStyle` Stata's further `passthru`, `mz`, and `split` suboptions of `gmm()/iv()` are not implemented.

IFE-GMM convergence Not guaranteed from an inconsistent starting point (Section 4.6).

IFE-GMM point-estimate bias Present without `bias_correction=True` (Section 4.9).

Unbalanced panels: attrition Interior gaps and late-starting units are cross-checked against a real Stata run to 6 decimal places (Section 5.1); end-of-panel attrition is validated internally only (no exception; finite inference across all three AR-test weighting branches) but not yet against an independent Stata run for that specific case.

IFE-GMM imputation under heavy missingness Missing panel cells feeding the defactoring loop (Section 4.5) are filled from the model's own evolving rank-r reconstruction – reliable for light-to-moderate missingness, but not independently stress-tested for panels with a large missing fraction, where this circularity could compound the convergence caveat of Section 4.6.

7 Conclusion

`pyxtabond2` reproduces Stata's `xtabond2` matrix algebra and diagnostics to the precision demonstrated in Section 5, with every remaining discrepancy from Stata specifically isolated to two documented, narrow regimes (Section 6) rather than left as an unexamined risk. Its original IFE-GMM extension is grounded, term for term, in Bai [2009]'s interactive-fixed-effects model, Hong et al. [2022]'s profile-GMM generalization to endogenous regressors, and Dhaene and Jochmans [2015]'s bias correction, with the two places its practical implementation departs from the cleanest available theory (the starting point for iteration; the default of `bias_correction=False`) documented as open questions for the user to weigh, not settled by fiat. Together with the companion User Manual, this note is intended to let a user or reviewer verify any single reported number back to its exact econometric source.

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